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Computer Architecture and Assembly Language Programming

Provided by Sarim

Q1) Define LOADS

Answer:

A load store architecture is an instruction set architecture that divides instructions into two categories

Memory access (load and store between memory and registers) and ALU operations (which only occur between registers).

Q3) Explain the purpose of using LDS instruction in a program.

Answer:

There are two special instructions that load a segment register and a general-purpose register from two consecutive memory locations.

LES loads ES while LDS loads DS.

Both these instructions have two parameters, one is the general-purpose register to be loaded and the other is the memory location from which to load these registers.

Q4) Write the name of four registers used in iAPX88 that can hold the address of the data.

Answer:

Name of four registers used in iAPX88 that can hold the address of the data.

BX, BP, SI, DI

Q5) Differences Between Near and Short Jump.

Answer:

A short JMP is the relative JMP that you refer to. It is encoded as two bytes; the actual JMP and the number of bytes +/- relative to the current IP.

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A near jump allows you to jump within the current "segment" (using real mode terms) or within the currently selected memory area in the CS selector.

Q6) Write an assembly language program for multiplication of two 4 bit numbers.

Answer:

```
[org 0x100]
;a:=b
mov eax, [num1]
mov ebx, [num2]
;a:=a+b
add eax, ebx
mov eax,0x4c00
int 0x21
```

Q7) Find Effective Address in Code.

Answer:

Effective address= Base address of CS register X 10_H + Address of IP

Q8) In 'Graphics Pixel' define the representation of BH, AL, CX and DX registers.

Answer:

In graphics programming, BH, AL, CX, and DX registers may represent various parameters such as color components, coordinates, or dimensions.

For example, BH might represent the blue component of a pixel color, AL could be used for storing individual pixel values, and CX and DX might hold coordinates for drawing operations.

Q9) Name of registers for service number and file attributes, used in Create/Truncate file using INT 21.

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Answer:

INT 21h (DOS Interrupt), the registers AH and AL are used for service numbers and file attributes, respectively, when performing file operations such as creating or truncating files.

Q8) Write the code to hook INT 4 of the processor by changing its vector with the address of a new routine "inthook"

Note: you are just required to write the code for interrupt hooking. No further details are required.

Answer:

```
[org 0x100]
Mov ax,0
Mov ex, ax;
Mov word[es:0], myisr
Mov [es:2], cs
Mov ax, 1000
Mov bl, 2
Div bl
mov ax 0x4c00
Int 0x21
```

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Q9) Define Topics Jumps, Flags Registers.

Answer:

Flags

The FLAGS register is the status register that contains the current state of a CPU. The size and meanings of the flag bits are architecture dependent.

Jumps

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The jump instruction transfers the program sequence to the memory address given in the operand based on the specified flag.

Q10) Show Illustration Ret or Call.

Answer:

The CALL instruction interrupts the flow of a program by passing control to an internal or external subroutine.

The RETURN instruction returns control from a subroutine back to the calling program and optionally returns a value

Q11) Write ASCII Code, A ascii code is 65 find the a ASCII code?

Answer:

a is 91

Q12) Write alternative instructions for CF, JBZ with character.

Answer:

For CF=1

JC= Jump if carry

JB= Jump if below

JNAE= Jump if not above or equal

Q13) Write a program square of 3 accumulator terminates 2 times.

Answer:

[org 0x0100]

mov bx,3

mov cx,3

mov ax,0

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mov ax, 0x4c00

int 0x21

total: dw 0

Q14) Find background and foreground color attributes and their physical address.

Answer:

Attributes for background and foreground color attributes byte RGB is

Foreground use 16 colors

Background use 8 colors

Q15) Difference between shift logical right and shift arithmetic right?

Answer:

Shift Arithmetic Right

A signed number holds the sign in its most significant bit. If this bit was one a logical right shifting will change the sign of this number because of insertion of a zero from the left.

Shift Logical Right

The shift logical right operation inserts a zero from the left and moves every bit one position to the right and copies the rightmost bit in the carry flag.

Q16) How MOVS instruction works?

Answer:

The mov instruction **copies the data item referred to by its second operand** (i.e. register contents, memory contents, or a constant value) into the location referred to by

its first operand (i.e. a register or memory). While register-to-register moves are possible, direct memory-to-memory moves are not.

Q17) MOV [AX+BC+30]

Find which physical strategy use

Answer:

The mode of addressing is known as **register indirect mode**. In this addressing mode, the offset address of data is in either BX or SI or DI Register. The default segment is either DS or ES. Example: MOV AX, [BX].

Q18) Purpose of "define byte"

Answer:

The purpose of "define byte" in assembly language is to allocate memory for a byte-sized variable or data item

Q19) Effective address

Answer:

Effective address refers to the memory address that is calculated and used to access data or operands in an instruction, taking into account any addressing modes or displacement

Q20) Physical address

Answer:

Physical address is the actual memory address in the physical memory hardware where data is stored or retrieved

Q21) XOR and AND gate

Answer:

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XOR and AND gates are fundamental logic gates used in digital electronics. XOR gate performs exclusive OR operation while AND gate performs logical AND operation

Q22) What are the 3 illegal addressing modes

Answer:

The three illegal addressing modes in assembly language depend on the architecture, but commonly include addressing modes that are not supported or not allowed by the processor, such as immediate addressing, indexed addressing without an index register, and absolute addressing without a base address

Q23) Write instruction for A on blue background and write code to set es for video memory

Answer:

```
MOV AH, 09h        ; Function to display string
MOV AL, 41h        ; ASCII code for 'A'
MOV BL, 01h        ; Video attribute for blue background
MOV BH, 00h        ; Page number (0 for default)
MOV CX, 01h        ; Number of characters to display (1)
MOV DX, 0007h      ; Video page and attribute (blue background)
INT 10h            ; Call BIOS interrupt to display character with attribute
MOV AX, 0B800h     ; Start address of video memory (color text mode)
MOV ES, AX         ; Set ES to point to video memory segment
```

Q24) What is purpose of Instructions Pointer Register?

Answer:

The purpose of the Instruction Pointer (IP) register is to store the address of the next instruction to be executed in the program sequence

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Q25) SS:SP

Answer:

SS:SP measure the size of the stack segment and the stack pointer within it, indicating the current top of the stack

Q25) Data bus control bus aur address bus thy sath memory aur processor to hmy directions batni thi

Answer:

Data bus carries data between the processor and memory or peripherals, the address bus specifies the memory location or I/O port being accessed, and the control bus carries control signals to coordinate data transfer between the processor and memory or I/O devices

Q26) Difference between shift logical right and shift arithmetic right?

Answer:

The main difference between shift logical right and shift arithmetic right is that in shift logical right, zeros are shifted into the vacant bit positions from the left, whereas in shift arithmetic right, the sign bit (MSB) is replicated to fill the vacant bit positions from the left.

Q27) How MOVS instructions works?

Answer:

The MOVS instruction in assembly language is used to move a block of data from one location to another. It works by copying the data from the source memory location to the destination memory location, typically used for moving strings or arrays of data.

Q28) Instruction MOV [AX+BX+30] iska btana tha k is m konsa physical design ya strategy use hogi?

Answer:

The instruction MOV [AX+BX+30] indicates a memory access operation where the physical addressing mode is used.

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The processor calculates the physical memory address by adding the contents of AX, BX, and the immediate value 30, and then retrieves or stores the data at that memory location.

Q29) Multiplication of 2 4 bits number

Answer:

Program for multiplication of two 4-bit numbers involves using iterative addition or bitwise shifting to perform the multiplication operation.

Q30) Background and foreground colors k attributes

Answer:

```
MOV AH, 09h       ; Function to display string
MOV AL, 'A'       ; ASCII code for 'A'
MOV BL, 1Eh       ; Attribute byte for blue background and white foreground
MOV BH, 00h       ; Page number (0 for default)
MOV CX, 01h       ; Number of characters to display (1)
MOV DX, 0007h     ; Video page and attribute (blue background, white foreground)
INT 10h           ; Call BIOS interrupt to display character with attribute
```

Q31) How instruction in carry flag is used as implied operand?

Answer:

Instruction in carry flag is used as an implied operand by checking the carry flag status after an arithmetic operation to determine if there was a carry or borrow during the operation.

Q32) Write down names of four registers in iapx88 that hold the address of data

Answer:

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Four registers in iapx88 that hold the address of data are IP (Instruction Pointer), SP (Stack Pointer), BP (Base Pointer), and DI (Destination Index).

Q33) Write a program in assembly language that clears the computer screen

Answer:

```
mov ah, 06h
mov al, 20h   ; ASCII code for space character
mov bh, 0
mov cx, 0
mov dh, 24
mov dl, 79
; mov bh, 07h  ; Example: white text on black background
int 10h
mov ah, 4Ch   ; Set exit function code
int 21h       ; Call DOS interrupt
```

Q34) How immediate operand is transferred into segment register? Explain with an example.

Answer:

Immediate operand can be transferred into a segment register by loading the immediate value into a general-purpose register and then using an appropriate instruction to move the value from the general-purpose register into the segment register

Q34) How many types of register which type is used lower to higher levels

Answer:

Several types of registers in a computer system, including data registers, address registers, control registers, and status registers.

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Registers used at lower levels include data registers and control registers, while those used at higher levels include address registers and status registers.

Q35) Why is the most important register used in assembly architecture

Answer:

The most important register used in assembly language programming is often considered to be the program counter (PC) or instruction pointer (IP), as it keeps track of the memory address of the next instruction to be executed, essential for program flow control.

Q36) Write procedure to clear any selective bit

Answer:

```
mov ax, 0b1101      ; Value in AX register: 1101 (binary)
mov cx, 2           ; Clear the 3rd bit (0-indexed)
call clear_bit      ; Call the procedure
```

Q37) In case of branching techniques used in Intel's architecture, which instruction is used in temporary diversion and how?

Answer:

In case of branching techniques used in Intel's architecture, the **JMP** (Jump) instruction is used for temporary diversion. It is used to transfer control unconditionally to a specified memory location.

Q38) Calculate physical memory address

a. 0010:0200

b. 0100:DFDA

Answer:

a. Physical address = (Segment * 16) + Offset = (0010 * 16) + 0200 = 0x210

b. Physical address = (Segment * 16) + Offset = (0100 * 16) + DFDA = 0x10FDA

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Q39) Mov (si),ax which instruction mode is used?

Answer:

The instruction mov (si), ax uses the Register Indirect addressing mode.

Q40) Why do we use ADC instead of Simple Add??

Answer:

We use ADC (Add with Carry) instead of simple ADD when we need to perform addition of two numbers along with the carry flag. ADC adds the source operand, the destination operand, and the carry flag.

Q41) All components of flag register:

. overflow flag

. zero flag

. parity flag

. signed flag

. carry flag

Answer:

Components of the flag register

- Overflow flag (OF)
- Zero flag (ZF)
- Parity flag (PF)
- Signed flag (SF)
- Carry flag (CF)

Q42) Video k liya koi code likhna tha or bg clr black character clr green ma print krna tha... Is ka code likhna tha

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Answer:

```
mov ah, 0x0B    ; Video interrupt function to set background and foreground color
mov bh, 0x00    ; Page number
mov bl, 0x00    ; Attribute (black background)
mov cx, 80      ; Number of columns (screen width)
mov dx, 0x02    ; Starting position (row 2, column 0)
int 0x10        ; Video interrupt

mov ah, 0x09    ; Video interrupt function to print characters
mov al, 'H'     ; ASCII value of the character to print
mov bl, 0x02    ; Attribute (green foreground)
mov cx, 1       ; Number of times to print the character
int 0x10        ; Video interrupt
```

Q43) What is the maximum memory that 8088 processor can access, Give Mathematical explanation

Answer:

The maximum memory that an 8088 processor can access is 1 MB (2^20 bytes).

This is because the 8088 processor uses a 20-bit address bus, allowing it to address up to 2^20 memory locations.

Mathematically, 2^20 = 1,048,576 bytes, which is equivalent to 1 MB

Q44) What will be effect of executing REPE and REPNE on zero flag?

Answer:

The REPE (Repeat Equal) and REPNE (Repeat Not Equal) instructions affect the zero flag

- REPE: If the zero flag is set (ZF = 1), the REPE instruction will repeat the following string instruction until the ZF becomes 0 or the CX register becomes 0.

- REPNE: If the zero flag is clear (ZF = 0), the REPNE instruction will repeat the following string instruction until the ZF becomes 1 or the CX register becomes 0.

Q45) Define Divide overflow error?

Mov bl, ___

Mov ax, ___

Div bl

Answer:

Mov bl, 0x8432

Mov ax, 2

Div bl

Q46) Calculate the physical memory address generated by the following segment offset pairs.

1. 1280 : 0321

2. FFFF : 4872

3. A4FB : 0333

4. FFFF : AAAA

5. 1234 : 4336

Answer:

Calculating physical memory address:

a. Physical address = (Segment * 16) + Offset = (1280 * 16) + 0321 = 20480 + 321 = 20801

b. Physical address = (FFFF * 16) + 4872 = 65535 * 16 + 4872 = 1048560 + 4872 = 1053432

c. Physical address = (A4FB * 16) + 0333 = 42107 * 16 + 0333 = 673712 + 333 = 674045

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d. Physical address = (FFFF * 16) + AAAA = 65535 * 16 + 0xAAAA = 1048560 + 43690 = 1092250

e. Physical address = (1234 * 16) + 4336 = 46656 + 4336 = 50992

Q47) Why We use Add with carry instead of ADD instructions?

Answer:

We use ADD with Carry (ADC) instead of simple ADD instructions when performing addition operations involving carry flag. ADC adds two operands along with the carry flag, allowing for multi-byte additions.

Q48) NOT operation with example

Answer:

Example:

Perform NOT operation on AL register

```
mov al, 10101010b   ; AL = 0xAA (binary)
not al              ; NOT operation on AL
```

Q49) Selective bit inversion operation on lower nibble

Answer:

Selective bit inversion operation on lower nibble of AL register

```
mov al, 11110000b   ; AL = 0xF0 (binary)
mov bl, 00001111b   ; BL = 0x0F (binary)
and al, bl          ; Perform bitwise AND operation with BL
```

Q50) Interrupt operations

Answer:

Interrupt operations are used for communication between hardware devices and the CPU. They allow external devices to interrupt the CPU to request attention or service.

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Q51) Segment override prefix function

Answer:

The segment override prefix function is used to override the default segment register used by the memory access instruction. It allows specifying a different segment register to access memory.

Q52) Difference assembler & compiler

Answer:

The assembler converts assembly language code into machine code (binary executable instructions), while the compiler converts high-level language code (like C, C++) into machine code.

Q53) Why address bus call unidirectional?

Answer:

The address bus is unidirectional because it only carries addresses from the CPU to the memory or I/O devices. Data flows bidirectionally on data buses, while control signals (like read/write signals) also have specific directions.

Q54) What is meant of implied Operand with example

Answer:

Implied Operand refers to an operand that is implicitly understood by the instruction, rather than explicitly specified in the instruction itself.

Example of an implied operand is the Accumulator register (AX) in the instruction ADD AX, BX, where the operand AX is not explicitly mentioned but is implied to be the destination operand.

Q55) What will be effect of executing REPE and REPNE on zero flag

Answer:

- REPE: Continues to repeat the following string instruction while the zero flag is set (indicating equality).
- REPNE: Continues to repeat the following string instruction while the zero flag is clear (indicating inequality).

Q56) In assembly language define the "Base Register indirect + offset.

Answer:

Base Register Indirect + Offset addressing mode in assembly language refers to accessing memory using a base register (such as BX, BP) combined with an offset value.

For example, in the instruction MOV AX, [BX+10], the contents of the memory address calculated as BX + 10 are moved into the AX register.

Q57) Which of the following are not bitwise operator?AND,MOV,ADD,Mul,oR,Not, TEST,Xor,DIV,ADC.

Answer:

Among the given options, MOV, ADD, MUL, DIV, and ADC are not bitwise operators.

Q58) Different line of control bus behave differently. Briefly explain the different operation performed by these lines.

Answer:

- Address lines: Transmit memory addresses from the processor to memory or I/O devices.
- Data lines: Carry data between the processor and memory or I/O devices.
- Control lines: Control signals for operations like read, write, or interrupt handling.

Q59) calculate physical memory address generated by the following segment offset pairs.

i= 0010:0200,

ii= 0100:DFAD

Answer:

i. Physical address = (0010h × 16) + 0200h = 0320h

ii. Physical address = (0100h × 16) + DFADh = 01000h + DFADh = DEADh

Q60) Assembly language program first 5 odd number k dx register mein store krna tha.

Answer:

```
mov ax, var1   ; Move value of var1 into AX
add ax, var2   ; Add value of var2 to AX
sub ax, var3   ; Subtract value of var3 from AX
imul ax, 5     ; Multiply AX by 5
```

Q61) Why the default association of BP is with SS?

Answer:

BP (Base Pointer) is by default associated with the SS (Stack Segment) because it is commonly used as a base address pointer for accessing variables and parameters on the stack.

Q62) illustrate the mapping of two-dimensional coordinate system into the 1 dimensional video memory with the helping formula

Answer:

Mapping of a two-dimensional coordinate system into 1-dimensional video memory can be done using the formula:

Video_Memory_Address = (Row_Number × Number_of_Columns_per_Row + Column_Number) × Bytes_per_Character

Q63) What is the direction of the following operation in the data bus?

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.Read

.Write

Answer:

- Read: Data flows from memory or I/O devices to the processor.
- Write: Data flows from the processor to memory or I/O devices.

Q64) Assembly language program first 5 odd number k dx register mein store krna tha

Answer:

```
mov cx, 5      ; Set loop counter to 5
mov bx, 1      ; Initialize BX to first odd number
start:
mov dx, bx     ; Move current odd number to DX
add bx, 2      ; Increment BX to next odd number
loop start     ; Repeat loop until CX becomes zero
```

Q65) Define the following assembly language expression: DAT: db 100

Answer:

DAT: db 100 defines a data byte (db) named DAT with an initial value of 100

Q66) What are local variable, how they are generated, and what is the most suitable place to store them?

Answer:

Local variables are variables defined within a specific scope, such as within a function or block of code.

They are typically allocated on the stack and can only be accessed within the scope in which they are defined.

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Q67) Convert the logical expression into assembly language:

if (AX>BX>CX)&&(AX>BX<CX)

AX = 5

Answer:

```
cmp AX, BX     ; Compare AX and BX
jg greater     ; Jump if AX is greater than BX
cmp AX, CX     ; Compare AX and CX
jl less        ; Jump if AX is less than CX
greater:
mov AX, 5      ; Move 5 into AX
less:
```

Q68) data bus and memory cell

Answer:

The data bus is used to transfer data between the processor and memory or I/O devices.

A memory cell is a basic storage unit in memory that holds binary data, typically consisting of several bits organized into bytes or words

Q69) Explain how push and pop change the value of stack Pointer

Answer:

When push instruction is executed, the stack pointer (SP) is decremented by the size of the data being pushed onto the stack. Conversely, when pop instruction is executed, the stack pointer is incremented by the size of the data being popped off the stack

Q70) Differentiate between assembler and compiler

Answer:

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Assembler translates assembly language code into machine code specific to the target processor architecture, producing object files.

Compiler translates high-level programming languages into machine code, generating executable files directly.

Q71) Name of the Pin in Port 37A used to enable triggering of ACK pin?

Answer:

The Pin in Port 37A used to enable triggering of the ACK pin is typically referred to as the 'Enable' or 'EN' pin

Q72) Add R2 and R3 and store their value in R4.

Answer:

add R4, R2, R3

Q58) Store krna tha Flag with zero value in "Next" operand.

Answer:

Storing a flag with zero value in the "Next" operand typically involves using conditional jump instructions based on the result of a comparison or logical operation that sets the zero flag when the condition is true.

Q73) Describe PUSHF and POPF in detail.

Answer:

PUSHF instruction pushes the contents of the flags register (including status and control flags) onto the stack, while POPF instruction retrieves and loads the flags register from the top of the stack

Q74) Discuss the structure of GDT.

Answer:

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Global Descriptor Table (GDT) is a data structure used in x86 architecture to define memory segments and their attributes in protected mode. It contains segment descriptors, each describing a segment's base address, limit, access rights, and other attributes.

Q75) In any assembly programming, describe how IRQ and interrupt are used to update system time.

Answer:

IRQ (Interrupt Request) and interrupts are used in assembly programming to handle events that require immediate attention, such as updating system time.

When an interrupt occurs, the processor suspends its current execution and transfers control to an interrupt handler routine

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